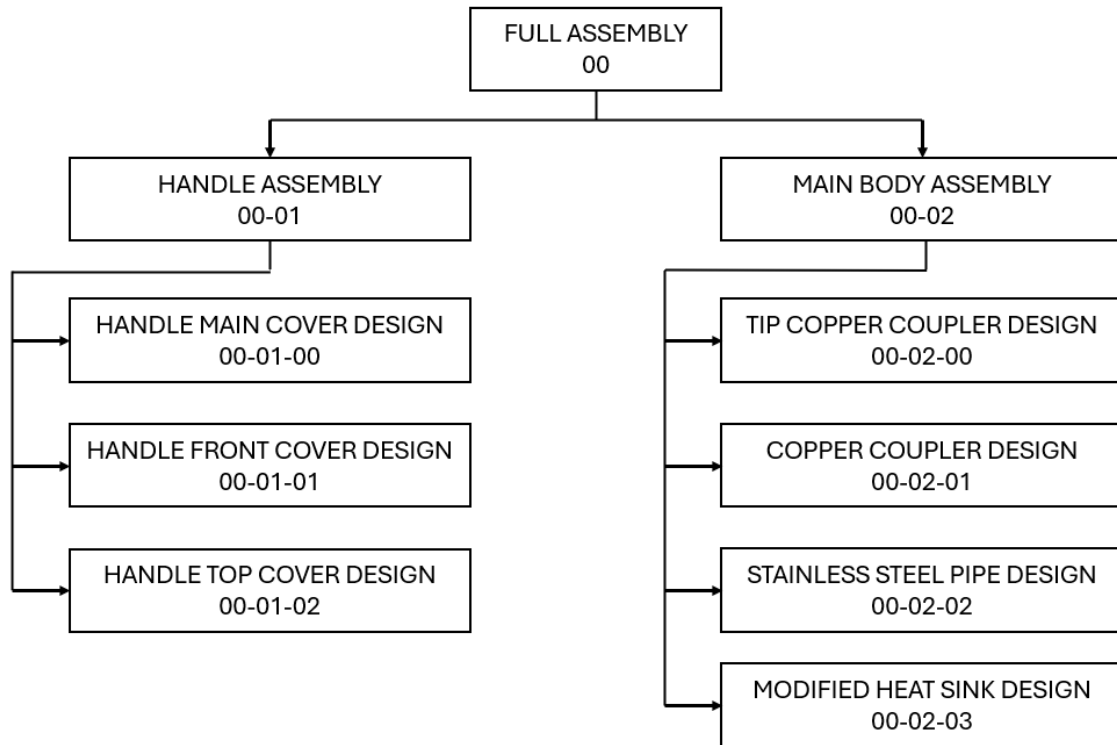


Laparoscope Drawing Pack

Student Number: 26122146

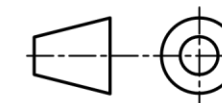
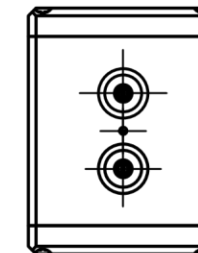
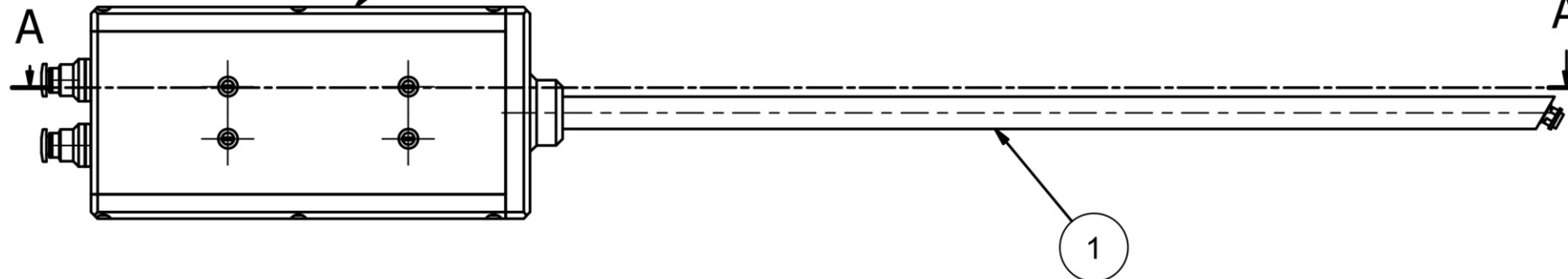
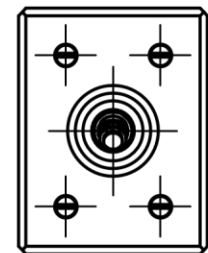
Student Initials and Surname: PW Breytenbach

Contact number: 078 780 9388

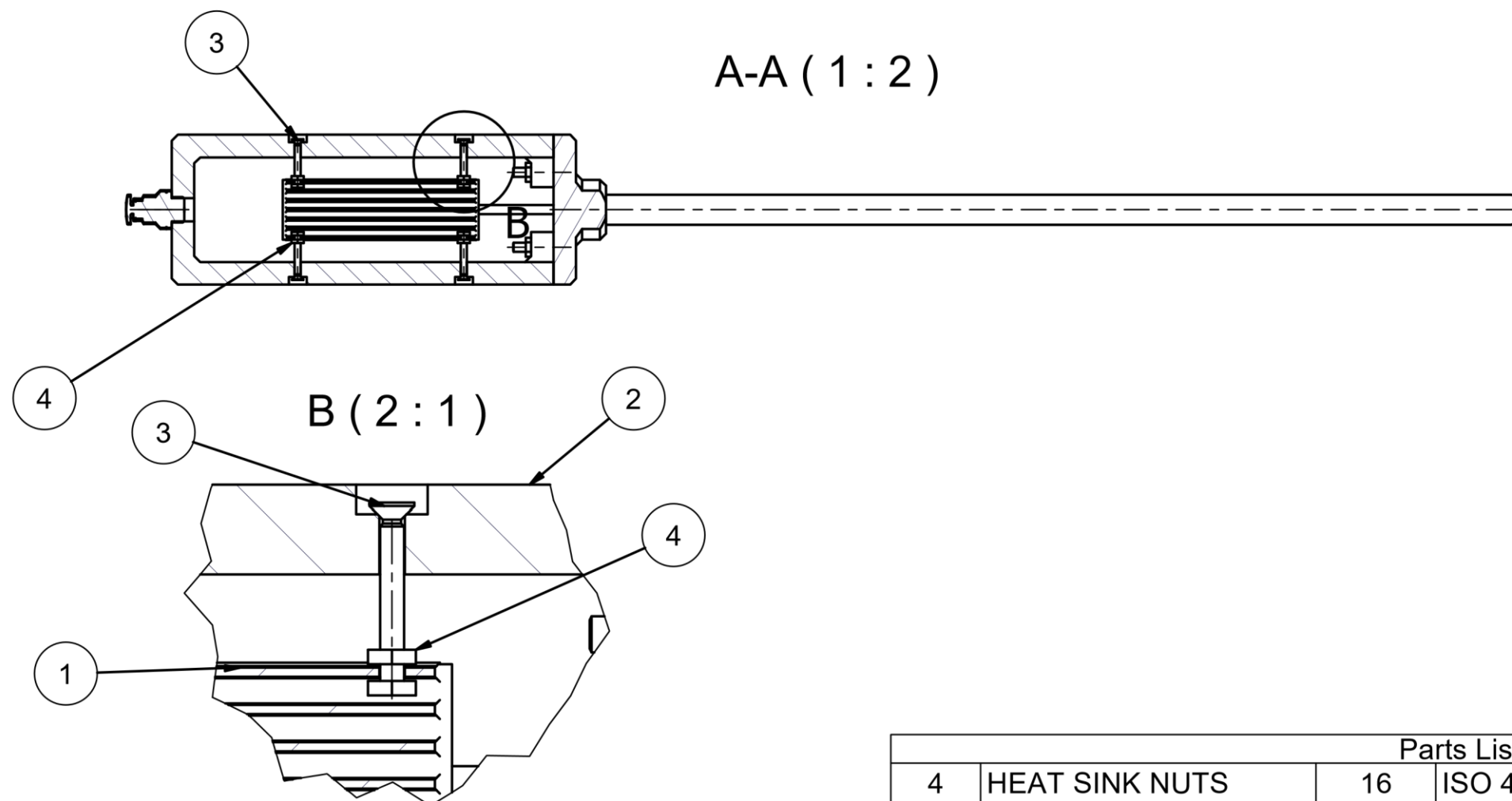


Notes for the assembly:

Parts that interface with each other that are not connected by a bolt and nut connection are secured by epoxy. Thus, all parts that interface with the heat pipes and stainless steel pipe will be secured by epoxy. The entire assembly of the design will be performed by the student, including the application of epoxy.



A-A (1 : 2)



UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK

PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY

STUDENT Nr. 26122146

DRAWN BY PW Breytenbach

CHECKED

Parts List			
4	HEAT SINK NUTS	16	ISO 4036 M2
3	HEAT SINK BOLTS	8	ISO 2009 M2 X 16 X 14
2	HANDLE ASSEMBLY	1	00-01
1	MAIN BODY ASSEMBLY	1	00-02
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS

SCALE ON A 3 1:4

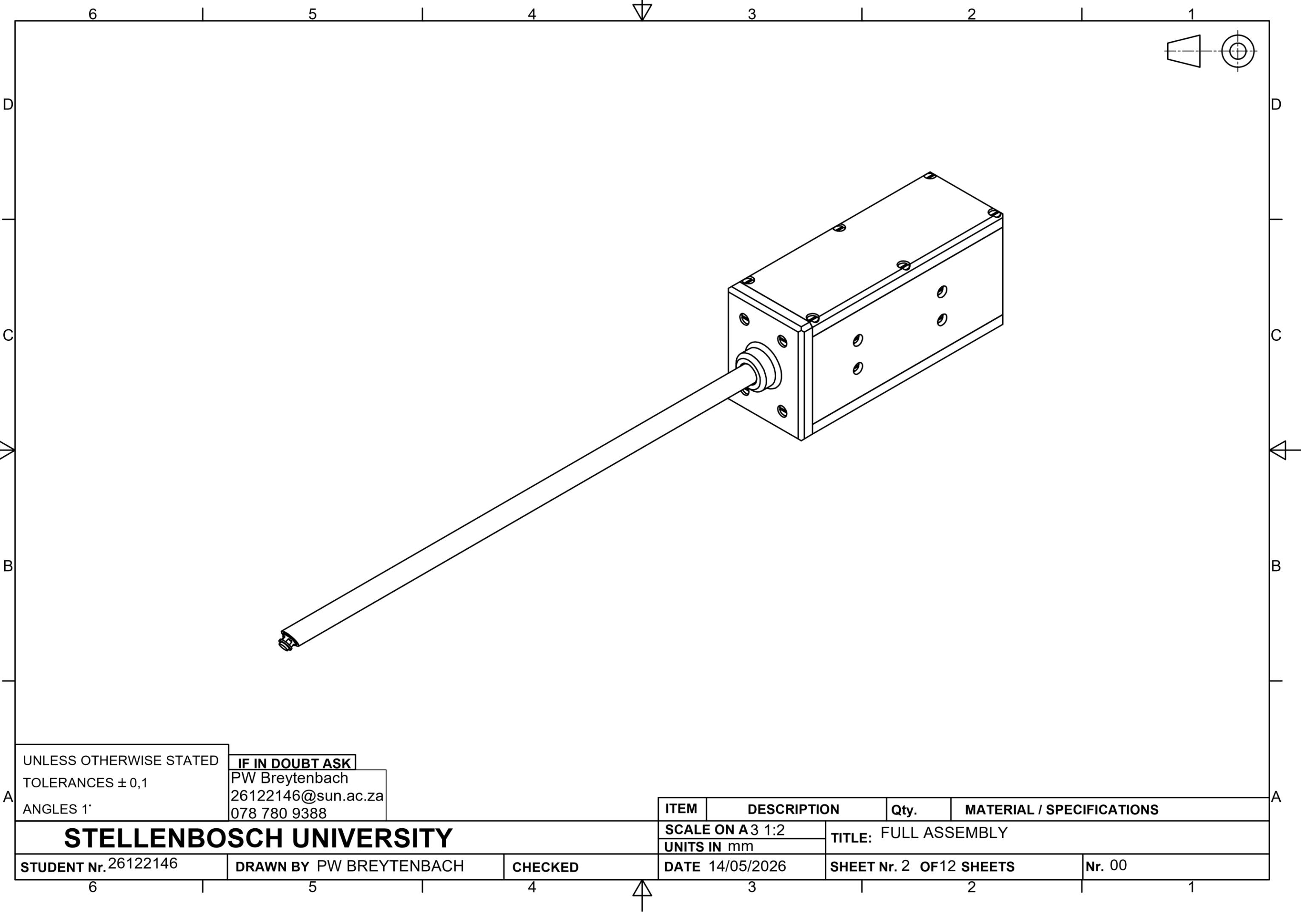
UNITS IN mm

TITLE: FULL ASSEMBLY

DATE	07/04/2026
------	------------

SHEET Nr. 1 OF SHEETS 12

Nr. 00

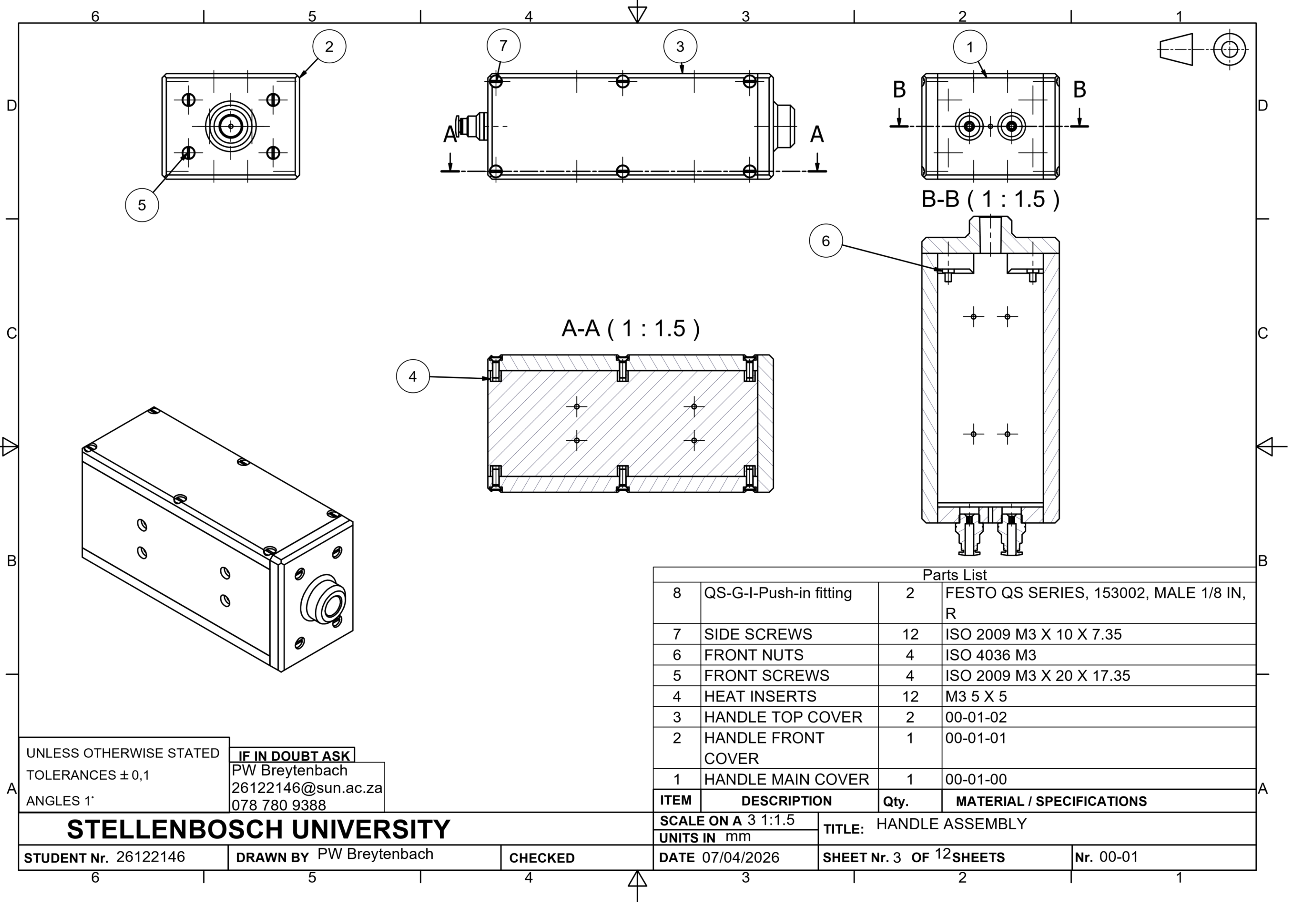


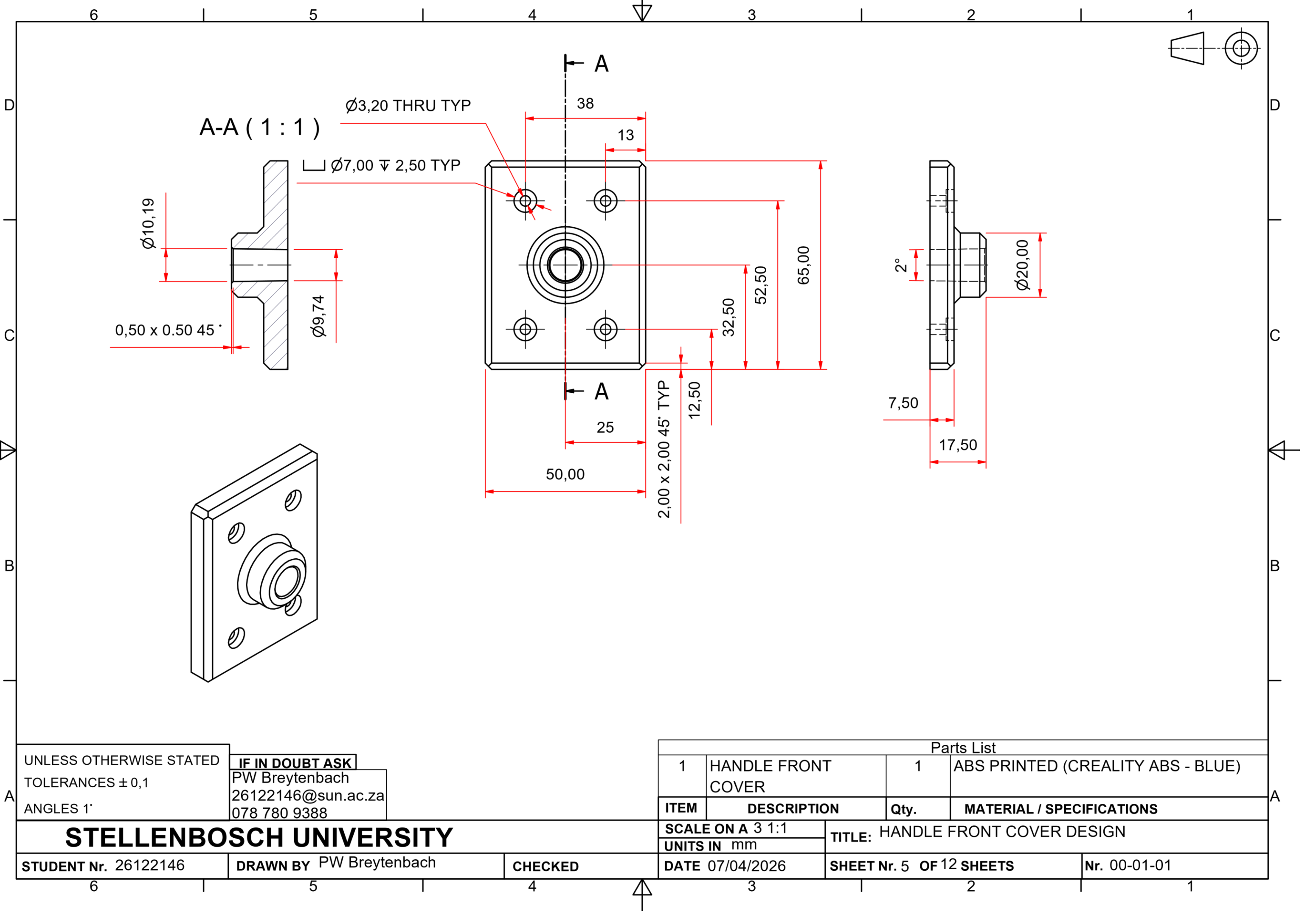
UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY			
STUDENT Nr. 26122146	DRAWN BY PW BREYTENBACH	CHECKED	

ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A3 1:2		TITLE: FULL ASSEMBLY	
UNITS IN mm			
DATE 14/05/2026	SHEET Nr. 2 OF12 SHEETS		Nr. 00



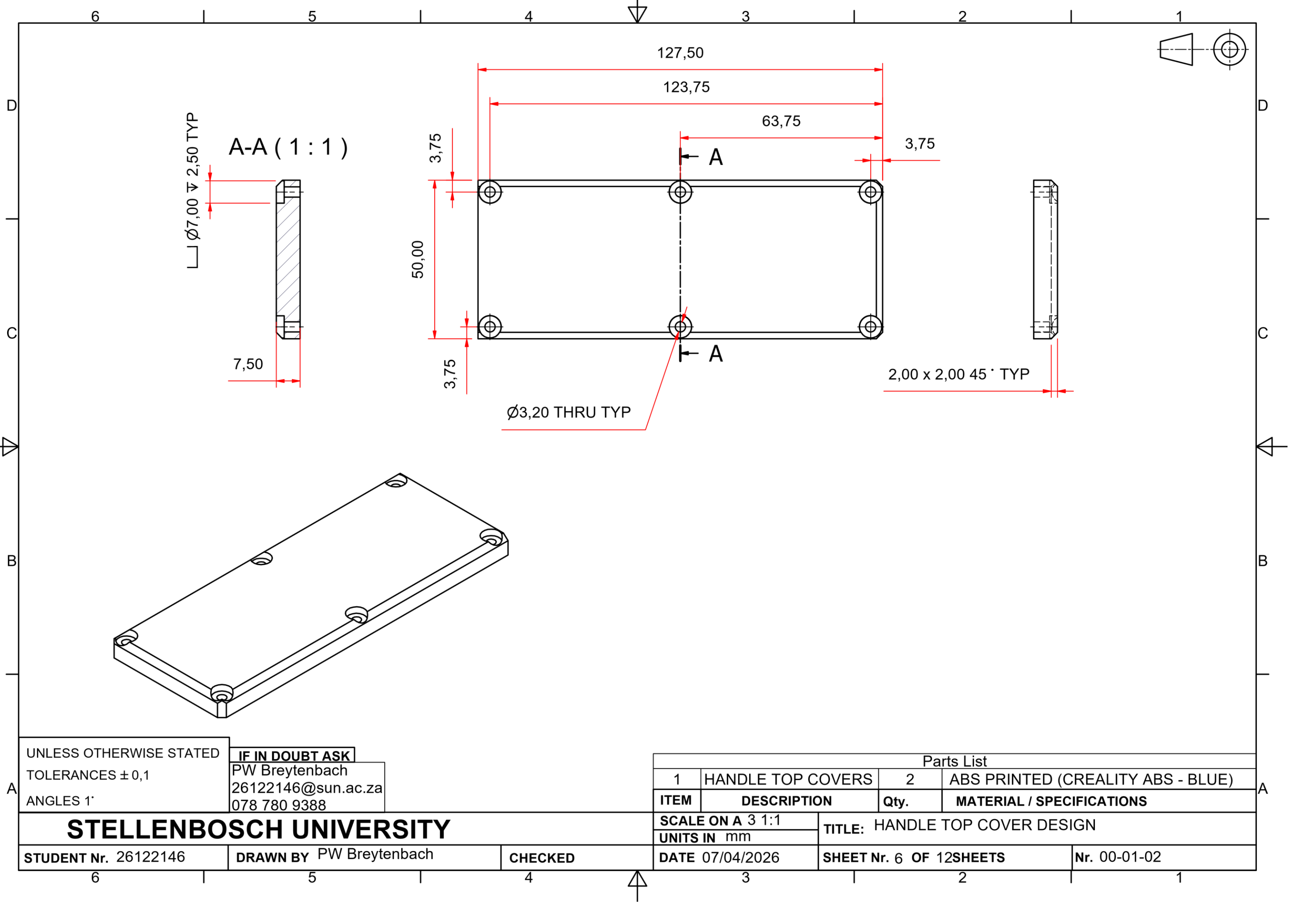


UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY		
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED

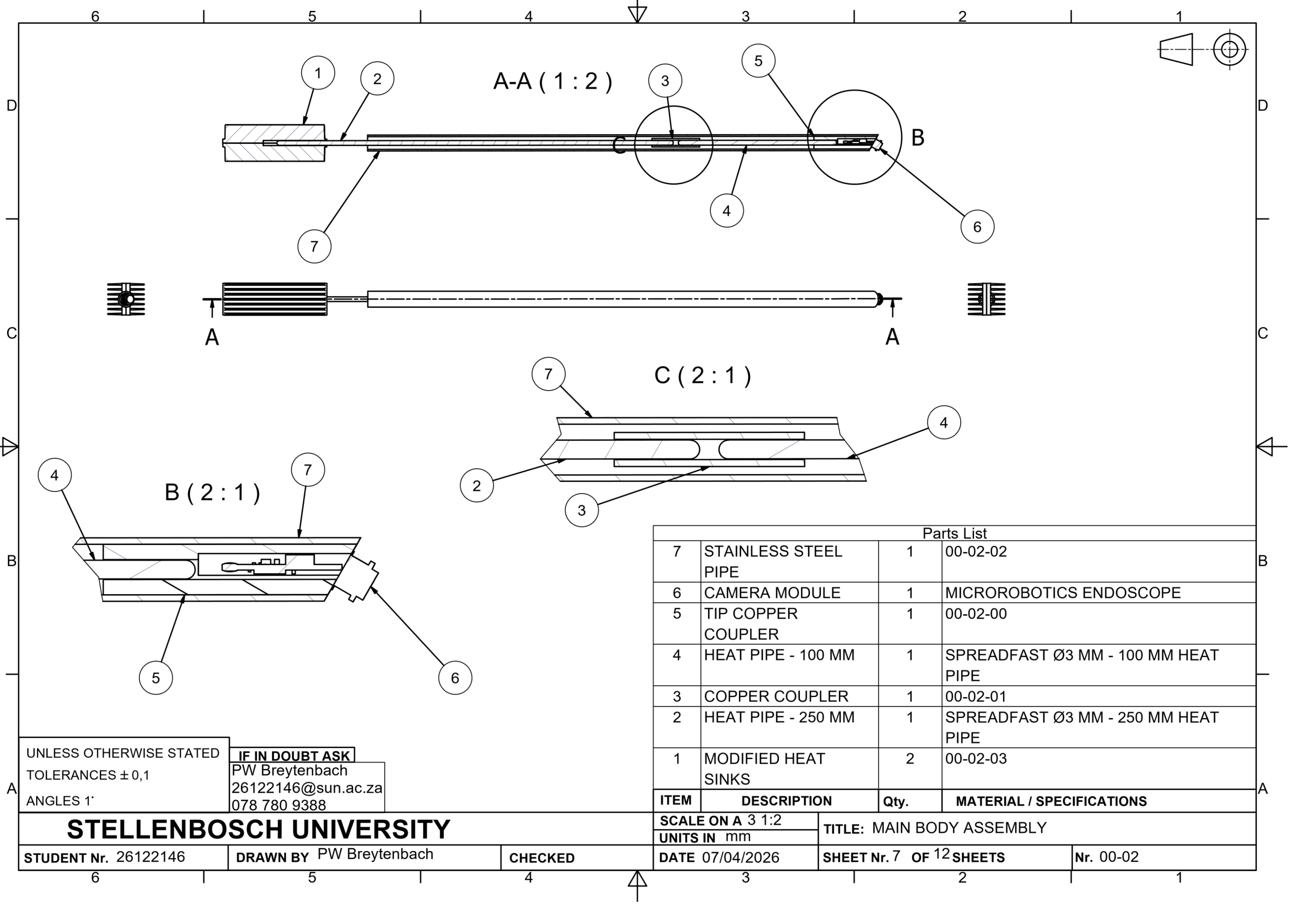
Parts List			
1	HANDLE FRONT COVER	1	ABS PRINTED (CREALITY ABS - BLUE)
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 1:1		TITLE: HANDLE FRONT COVER DESIGN	
UNITS IN mm			
DATE 07/04/2026		SHEET Nr. 5 OF 12 SHEETS	Nr. 00-01-01



UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

Parts List			
1	HANDLE TOP COVERS	2	ABS PRINTED (CREALITY ABS - BLUE)
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 1:1 UNITS IN mm		TITLE: HANDLE TOP COVER DESIGN	
STUDENT Nr. 26122146		DRAWN BY PW Breytenbach	
CHECKED		DATE 07/04/2026	
		SHEET Nr. 6 OF 12SHEETS	
		Nr. 00-01-02	



UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY

STUDENT Nr. 26122146

DRAWN BY PW Breytenbach

CHECKED

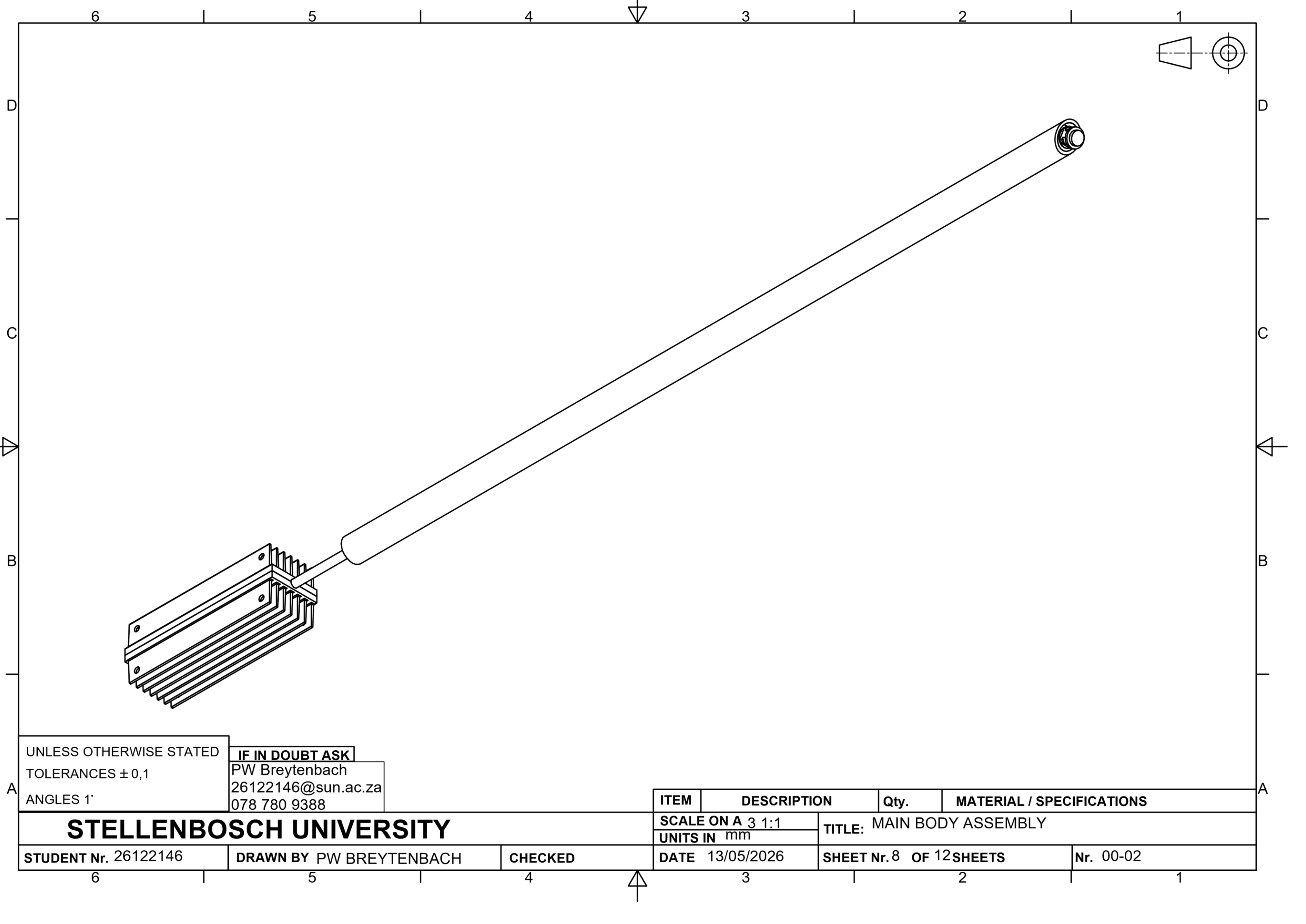
SCALE ON A 3 1:2
UNITS IN mm

DATE 07/04/2026

TITLE: MAIN BODY ASSEMBLY

SHEET Nr. 7 OF 12 SHEETS

Nr. 00-02

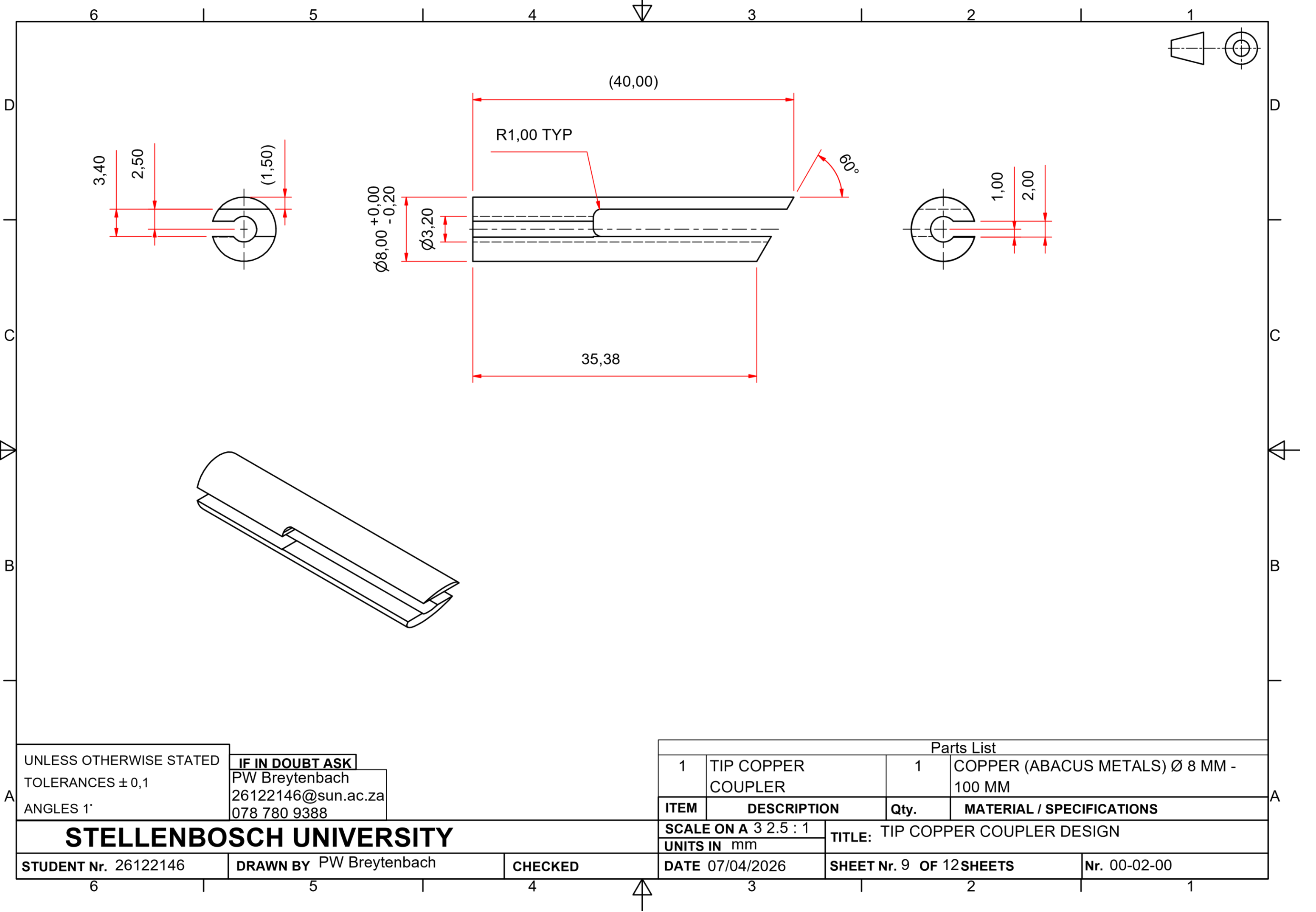


UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY			
STUDENT Nr. 26122146	DRAWN BY PW BREYTENBACH	CHECKED	

ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS	
SCALE ON A 3 1:1		TITLE: MAIN BODY ASSEMBLY		
UNITS IN mm				
DATE 13/05/2026		SHEET Nr.8 OF 12SHEETS		Nr. 00-02

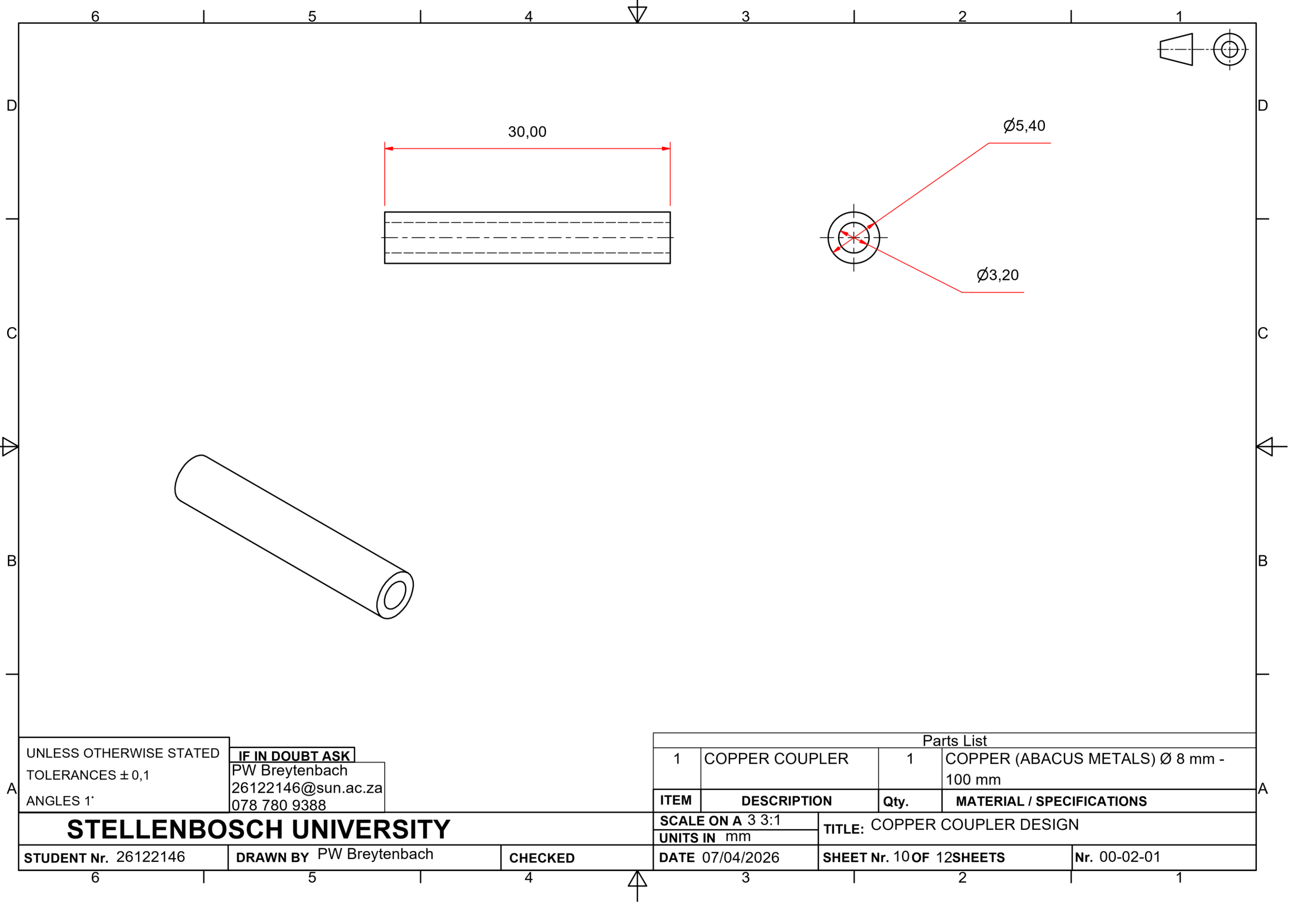


UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY		
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED

Parts List			
1	TIP COPPER COUPLER	1	COPPER (ABACUS METALS) Ø 8 MM - 100 MM
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 2.5 : 1 UNITS IN mm		TITLE: TIP COPPER COUPLER DESIGN	
DATE 07/04/2026		SHEET Nr. 9 OF 12 SHEETS	
		Nr. 00-02-00	

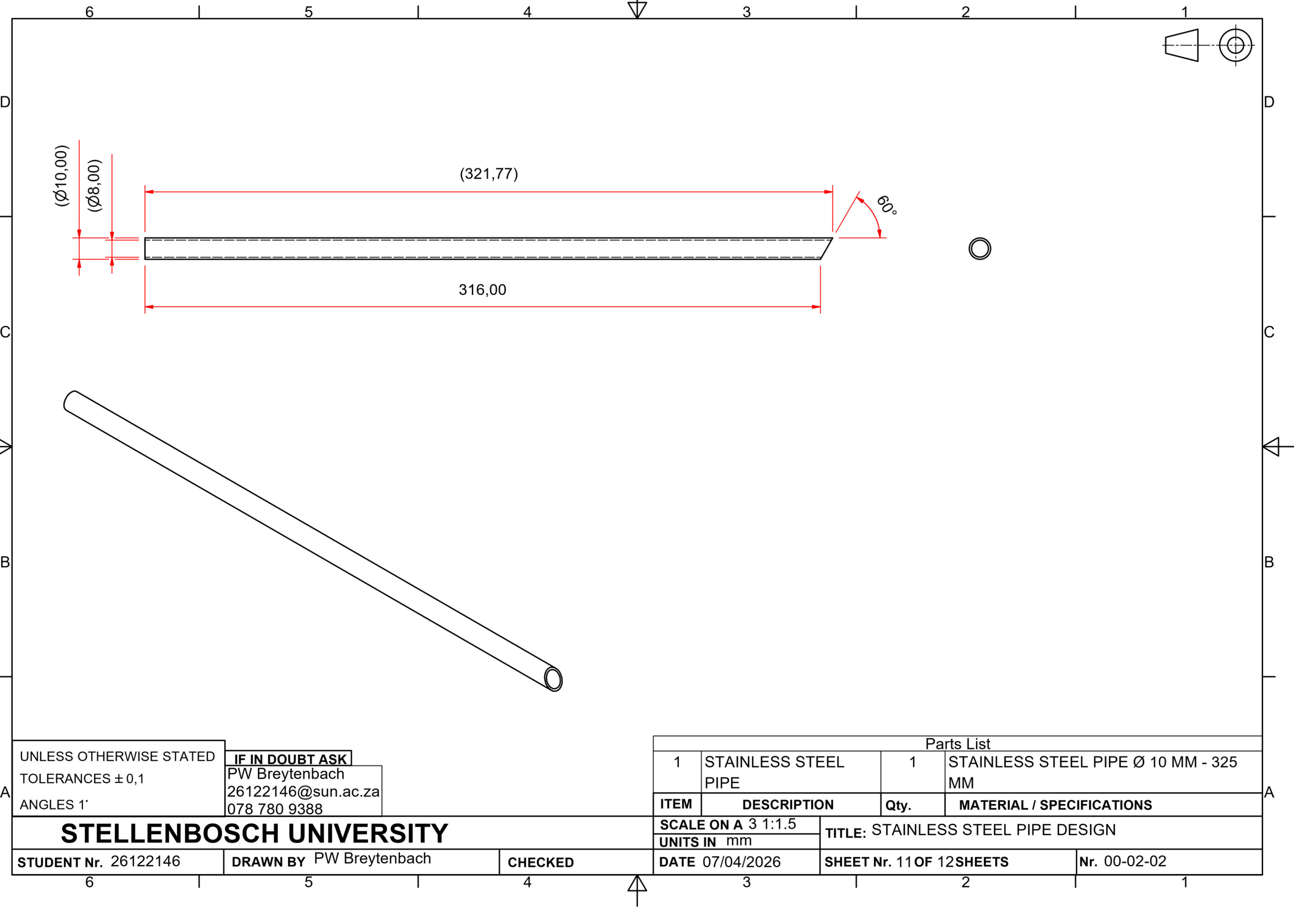


UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY		
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED

Parts List			
1	COPPER COUPLER	1	COPPER (ABACUS METALS) Ø 8 mm - 100 mm
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 3:1 UNITS IN mm		TITLE: COPPER COUPLER DESIGN	
DATE 07/04/2026		SHEET Nr. 10 OF 12 SHEETS	Nr. 00-02-01

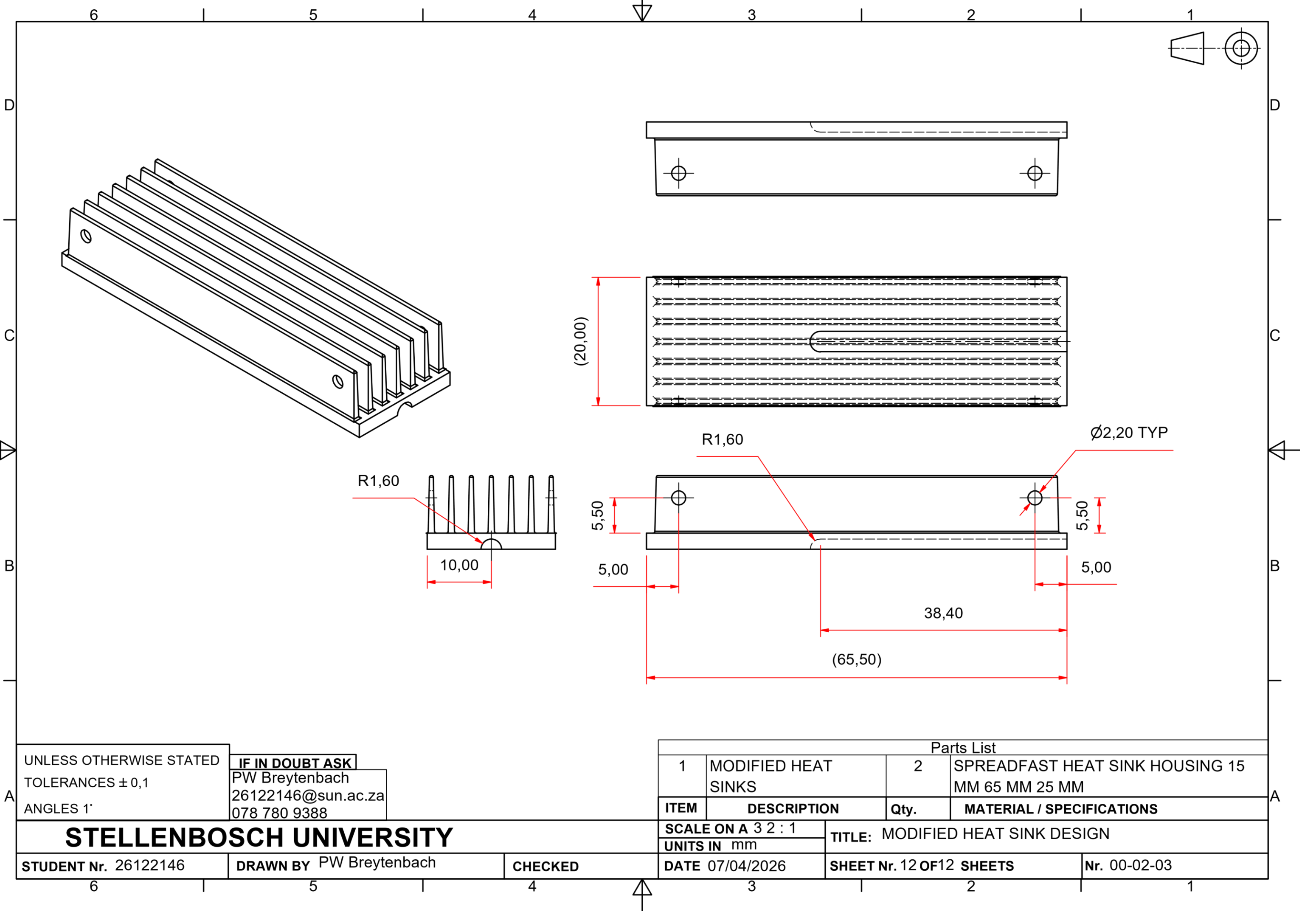


UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

STELLENBOSCH UNIVERSITY		
STUDENT Nr. 26122146	DRAWN BY PW Breytenbach	CHECKED

Parts List			
1	STAINLESS STEEL PIPE	1	STAINLESS STEEL PIPE Ø 10 MM - 325 MM
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 1:1.5		TITLE: STAINLESS STEEL PIPE DESIGN	
UNITS IN mm			
DATE 07/04/2026		SHEET Nr. 11 OF 12SHEETS	Nr. 00-02-02



UNLESS OTHERWISE STATED
TOLERANCES $\pm 0,1$
ANGLES 1°

IF IN DOUBT ASK
PW Breytenbach
26122146@sun.ac.za
078 780 9388

Parts List			
1	MODIFIED HEAT SINKS	2	SPREADFAST HEAT SINK HOUSING 15 MM 65 MM 25 MM
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
SCALE ON A 3 2 : 1 UNITS IN mm		TITLE: MODIFIED HEAT SINK DESIGN	
STUDENT Nr. 26122146		DRAWN BY PW Breytenbach	
CHECKED		DATE 07/04/2026	
		SHEET Nr. 12 OF 12 SHEETS	
		Nr. 00-02-03	